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Field of Testing : Chemical Testing and
Microbiological Testing

Chemical testing

Products	Specific Test	Method/Reference Standard
I. Foods		
01. Cereal and Cereal Products		
01.1 Breakfast cereals 01.2 Cereal/cereal grains 01.3 Cultured seeds and grains 01.4 Soya flours concentrates and isolates 01.5 Flour, corn meal, corn grits, semolina 01.6 Frozen entrees containing rice or corn flour	Moisture	Gravimetric/ AOAC 22 ND edition
01.7 Soy protein 01.8 Tofu 01.9 Pasta products and noodles (e.g. rice paper, rice vermicelli, soybean pastas and noodles) 01.10 Starch	Ash	Gravimetric/ AOAC 22 ND edition
02. Nuts and nut products		
02.1 Peanut butter and other nut butters	Moisture	Gravimetric/ AOAC 22 ND edition
	Ash	Gravimetric/ AOAC 22 ND edition
03. Dairy products		
03.1 All cheese made from pasteurized milk (cottage cheese, soft & semi-solid cheese)		
03.2 Processed cheese spread		
03.3 Ice cream and sherbet plain and flavoured		

03.4 Ice cream with added ingredients	Moisture	Gravimetric/ AOAC 22 ND edition
03.5 Flavored ice		
03.6 Milk powders (whole, nonfat or filled milk, buttermilk, whey & whey protein concentrate)		
03.7 Sweetened Condensed milk	Ash	Gravimetric/ AOAC 22 ND edition
03.8 Liquid Milk (evaporated or Ready To Drink) and Cream (Ultra Heat Temperature/sterilized)		
03.9 Pasteurized milk		
03.10 Pasteurized cream		
03.11 Yogurt and other fermented milk		
04. Meat and Meat products		
04.1 Dried animal products	Moisture	Gravimetric/ AOAC 22 ND edition
04.2 Meat paste and pate’ (heat treated)		
04.3 Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat		
04.4 Packaged cooked cured/salted meat (ham, bacon)		
04.5 Fermented, comminuted meat, not cooked (dry and semi-dry fermented sausages)		
04.6 Cooked poultry meat, frozen to be re- heated before eating (e.g. prepared frozen meals)	Ash	Gravimetric/ AOAC 22 ND edition

05. Fish and fish products, including molluscs, crustaceans, and echinoderms		
05.1 Fresh and frozen fish and cold- smoked	Moisture	Gravimetric/ AOAC 22 ND edition
05.2 Pre-Cooked Breaded Fish		
05.3 Frozen cooked crustaceans		
05.4 Cooked, chilled & frozen crabmeat		
05.5 Fish and shellfish products in hermitically sealed containers	Ash	Gravimetric/ AOAC 22 ND edition
05.6 Smoked, dried, fermented, and /or salted fish and fish products		
06. Sugar and sugar products		
06.1 Refined and raw sugars	Moisture	Gravimetric/ AOAC 22 ND edition
06.2 Brown sugar		
06.3 Sugar solutions and syrups		
06.4 Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings		
06.5 Honey		

06.6 Table-top sweeteners, including those containing high-intensity sweeteners	Ash	Gravimetric/ AOAC 22 ND edition
08. Fruits, jams and other fruit products		
08.1 Frozen fruits 08.2 Coconut (desiccated) 08.3 Sun dried fruit 08.4 Jams, jellies, marmalades	Moisture	Gravimetric/ AOAC 22 ND edition
08.5 Fruit-based spreads 08.6 Candied fruit 08.7 Fruit preparations (pulp, purees, fruit toppings and coconut milk) 08.8 Fermented fruit products 08.9 Fruit fillings for pastries	Ash	Gravimetric/ AOAC 22 ND edition
12. Fruit juices, drinks and concentrates		
12.1 Fruit and vegetable juices 12.2 Fruit and vegetable nectars 12.3 Water-based flavoured drinks (Carbonated, Non- carbonated, Concentrates (liquid or solid))	Moisture	Gravimetric/ AOAC 22 ND edition
12.4 Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	Ash	Gravimetric/ AOAC 22 ND edition
16. Bakery Products		
16.1 Frozen bakery products (ready to eat) with low-acid or high a_w fillings or toppings 16.2 Frozen bakery products (to be cooked) with low-acid or high a_w fillings or toppings (eg. meat pies, pizza) 16.3 Frozen and refrigerated doughs (Chemically leavened)	Moisture	Gravimetric/ AOAC 22 ND edition
16.4 Frozen and refrigerated doughs 16.5 Baked goods 16.6 Coated or filled, dried shelf- stable biscuits	Ash	Gravimetric/ AOAC 22 ND edition
25. Food supplement and / or Dietary supplement	Moisture	Gravimetric/ AOAC 22 ND edition
	Ash	Gravimetric/ AOAC 22 ND edition

Microbiological Testing:

PRODUCTS	SPECIFIC TEST	METHOD/REFERENCE STANDARD	
01.0 Milk and Dairy Products			
01.1 Milk Powders 01.2 Sweetened Condensed Milk/ Creamer 01.3 Liquid Milk (evaporated or ready to drink) and cream (UHT/sterilized) 01.4 Pasteurized milk 01.5 Pasteurized cream 01.6 Yogurt and other fermented milk 01.7 Heat treated fermented milks (sterilized, UHT) 01.8 Processed cheese spread 01.9 Soft Cheese (from pasteurized milk) 01.10 Hard and semi-hard cheese 01.11 All raw milk cheese; raw milk unripened cheese with moisture > 50%, pH > 5.0 01.12 Cream cheese products	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method	
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04	
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03	
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12	
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18	
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21	
	02.0 Fats and oils, and fat emulsions		
	02.1 Butter (whipped, pasteurized) 02.2 Butter made from unpasteurized milk or unpasteurized milk products 02.3 Margarine 02.4 Virgin Coconut Oil	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
Coliforms and <i>E. coli</i>		Multiple Tube Fermentation Technique BAM Chapter 04	
Aerobic Plate Count		Conventional Plate Count Pour Plate Method – BAM Chapter 03	
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12	
Yeast and Mold Count		Conventional Plate Count Pour Plate Method – BAM Chapter 18	
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21	

03.0 Edible ices, including sherbet and sorbet		
03.1 Ice cream and sherbet (plain and flavoured) 03.2 Ice cream with added ingredient (nuts, fruits, cocoa etc.) 03.3 Flavored ice (e.g. Ice candy) 03.4 Ice Products (pre-packaged tubed and cubed ice)	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
	Heterotrophic Plate Count	SMEWW 24 th Edition Pour Plating Method Petrifilm AOAC:9215A6a
		SMEWW 24 th Edition Pour Plating Method 9000:9215A-9215B
04.0 Confectioneries		
04.1 Cocoa powder 04.2 Chocolate products 04.3 Chocolate confectioneries (chocolate bars, blocks and bonbon) 04.4 Sugar confectioneries (hard and soft candies, toffees, caramel, fondants, creams, nougats and pastes) 04.5 Ethnic Milk-based Confectioneries (Pastillas and Yema) 04.6 Ethnic Flour-based Confectioneries (Polvoron, Piaya and Barquillos) 04.7 Coconut Sap Sugar 04.8 Dry, Granulated Sugar 04.9. Liquid Sugar (sugar syrup) 04.10 Non-nutritive sweeteners (e.g. aspartame, sucralose) and bulk sweeteners (e.g. polysols)	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21

05.0 Fruits and vegetables, nuts and seeds		
05.1 Frozen vegetables and fruits (pH > 4.5) 05.2 Fermented fruits and vegetables, Ready to Eat (e.g. Kimchi) 05.3 Fruits and vegetable products in hermetically sealed container (thermally processed) 05.4 Dried vegetables 05.5 Coconut (desiccated) 05.6 Peanut Butter & other Nut Butters 05.7 Sun Dried Fruits 05.8 Dried Fruits (e.g. dried mangoes, dates, figs, apricots, raisins etc.) 05.9 Ethnic Food Products Sweet Preserves in Syrup e.g. Kaong, Langka, Banana, Legumes, Chickpeas, Red Beans, White Kidney Beans, Coconut, Nata de Coco, Mixed Preserves (halo-halo) 05.10 Purple Yam (Ube) Jam (Halaya) 05.11 Malunggay (Moringa) Powder	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21
06.0 Egg and egg products		
06.1 Pasteurized egg products (liquid, smoked, frozen or dried)	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21

07.0 Cereals and cereal products		
07.1 Breakfast cereals 07.2 Cereal/cereal grains 07.3 Cultured seeds and grains (Sprouted Seeds) e.g. bean sprouts, alfalfa etc. 07.4 Soya flours, concentrates and isolates 07.5 Flour, corn meal, corn grits, semolina 07.6 Dry Mixes (e.g. Pancake Mix, Purple Yam Mix) 07.7 Frozen entrees containing rice or corn flour as main ingredient 07.8 Soy protein 07.9 Tofu 07.10 Pasta products and noodles Uncooked (wet and dry 07.11 Starch 07.12 Coconut Flour 07.13 Instant Noodles (Noodle Cake) 07.14 Instant Noodles (mixed with seasoning) 07.15 Rice based desserts (rice cakes), ready-to-eat e.g. kakanin, suman, layered, moulded, rice balls, mochi	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21
08.0 Bakery Products		
08.1 Frozen bakery products (ready to eat) with low acid or high a _w fillings or toppings 08.2 Frozen bakery products (to be cooked) with low acid or high fillings or toppings (e.g. meat pies and pizzas) 08.3 Frozen and refrigerated doughs 08.4 Baked goods (e.g. breads, cakes, pie crust, pastries and fried doughs) 08.5 Coated or filled, dried shelf-stable biscuits	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21

09.0 Ready-to-eat savouries		
09.1 Snacks foods (e.g. potato chips, popcorn, pretzels, rice crackers, flavored crackers, processed whole nuts, yoghurt- cereal- and honey-covered nuts, dried fruit nut and cereal snacks, crackers with fish, fish product or flavoring)	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21

10.0 Meat and meat products		
10.1 Dried animal products 10.2 Meat paste and pate' (heat treated) 10.3 Cold cuts, frozen and chilled hotdogs, corned beef, luncheon meat 10.4 Packaged cooked cured/salted meat (ham, bacon) 10.5 Fermented, comminuted meat, not cooked (dry and semi-dry fermented sausages) 10.6 Cooked poultry meat, frozen to be re-heated before eating (e.g. prepared frozen meals chicken burgers, chicken turkey rolls, chicken nuggets, other breaded poultry meat products) 10.7 Cured/smoked poultry meat 10.8 Dehydrated poultry products 10.9 Meat products in hermetically sealed containers 10.10 Marinated Meat Products e.g. Marinated meat and meat preparations (tapa, sisig, etc.), Marinated Poultry, Dim Sum made from meat (siomai) 10.11 Minced meat and meat preparations made from poultry meat intended to be eaten cooked 10.12 Minced meat and meat preparations made from species other than poultry intended to be eaten cooked 10.13 Foods cooked immediately prior to sale or consumption e.g. Takeaway food, burgers, kebabs, sausages, pizza, ready meals (cook/chill and cook/freeze) after regeneration	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method – Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method – Petri film AOAC Chapter 17 997.02p 19-21

11.0 Fish and other seafood products		
11.1 Fresh frozen fish and cold smoked 11.2 Pre-cooked breaded and fish 11.3 Frozen raw crustaceans 11.4 Frozen cooked crustaceans 11.5 Cooked crustaceans 11.6 Cooked, chilled and frozen crabmeat 11.7 Fresh and frozen bivalve molluscs 11.8 Fish and shellfish products, cooked crustaceans in hermetically sealed containers (thermally processed) e.g. cooked bagoong/shrimp paste 11.9 Ethnic food products – Dried, salted fish 11.10 Smoked Fish 11.11 Salt Fermented Fish and Shrimps (bagoong) 11.12 Thermally processed fish products 11.13 Aquatic products – cooked (heat-treated) aquatic product; intended for human consumption aquatic plant products (algae) 11.14 Fish and crustacean based processed meat e.g. fish ball, squid ball	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method - Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method - Petri film AOAC Chapter 17 997.02p 19-21

12.0 Spices, soups, sauces, salads and protein products		
12.1 Dry mixes for soup and sauces 12.2 Yeast 12.3 Spices 12.4 Spices (ready to eat) 12.5 Food Grade Gelatin 12.6 Emulsified Sauce pH ≤4.6 (e.g. Mayonnaise, Thousand Island, Ranch, French) 12.7 Instant noodles seasoning 12.8 Ready to eat Flavourings/ Instant condiments (e.g. soy sauce and sauce and sauce products, composite seasonings) 12.9 Aquatic dressing 12.10 Salad and Sandwich spreads (excluding cocoa milk based sandwich spreads)	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method- Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method- Petri film AOAC Chapter 17 997.02p 19-21
13.0 Beverages		
13.1 Non-alcoholic beverages (e.g. Ready to drink softdrinks, iced tea, energy drinks, jelly drinks) 13.2 Frozen juice concentrate 13.3 Juices in hermetically sealed containers (Tetra Pack etc.) 13.4 Powdered beverages (e.g. iced tea, powdered juices /mixes) 13.5 Fruit beverage products 13.6 Bottled Water (Purified/Distilled Water) 13.7 Bottled Water (Mineral Water) 13.8 Chilled Young Coconut Water (Buko Juice)	<i>Salmonella sp.</i> Detection	Streak Plating Method / Rapid Detection of <i>Salmonella</i> Merck Method
	Coliforms and <i>E. coli</i>	Multiple Tube Fermentation Technique BAM Chapter 04
	Aerobic Plate Count	Conventional Plate Count Pour Plate Method – BAM Chapter 03
		Pour Plating Method - Petri film AOAC Chapter 17 990.12 p12
	Yeast and Mold Count	Conventional Plate Count Pour Plate Method – BAM Chapter 18
		Pour Plating Method - Petri film AOAC Chapter 17 997.02p 19-21
	Heterotrophic Plate Count	SMEWW 24 th Edition Pour Plating Method Petrifilm AOAC:9215A6a
		SMEWW 24 th Edition Pour Plating Method 9000:9215A-9215B

Legend to Reference Methods

AOAC - Association of Official Analytical Collaboration International

BAM - Bacteriological Analytical Manual

SMEWW – Standard Methods for the Examination of Water and Wastewater

***** *Nothing Follows* *****